

Dictionaries (Exercises)

1. `d = {1:'i', 2:'ii', 3:'iii'}`
`d` `{1: 'i', 2: 'ii', 3: 'iii'}`
2. `d[2]`
3. `d[3] = 'many'`
`d[3]`
4. `d[2] = 'many'`
`d`
5. `3 in d`
6. `len(d)`
7. `del d[2]`
`d`
8. `{}`
9. `type({})`
10. `d = {1:'a', 2:'b', 3:'c'}`
`[d[n] for n in d]`

Dictionaries

A *dictionary* (of type `dict`) stores a set of *keys* and associated *values*. The analogy is to a dictionary for a natural human language, in which the keys are words and the values are their definitions.

Each key and its value are separated by a colon. The entire dictionary is delimited by curly braces `{` and `}`.

Pairs can be looked up or replaced using square brackets, as with a list. With a dict, the square brackets contain the key in question rather than an index.

The `in` operator indicates if a given key is present. `len` returns the size of the dict.

To remove a pair from a dict, use the `del` operator:

```
del dictionary[key]
```

.

`[]` is obviously the empty list, but is `{}` an empty set or an empty dict? It turns out it's an empty dict. The empty set is represented as `set()`.

The keys of a dictionary can be

Dictionaries (Solutions)

1. `d = {1:'i', 2:'ii', 3:'iii'}`
`d` `{1: 'i', 2: 'ii', 3: 'iii'}`
2. `d[2]` `'ii'`
3. `d[3] = 'many'`
`d[3]` `'many'`
4. `d[2] = 'many'`
`d` `{1: 'i', 2: 'many', 3: 'many'}`
5. `3 in d` `True`
6. `len(d)` `3`
7. `del d[2]`
`d` `{1: 'i', 3: 'many'}`
8. `{}` `{}`
9. `type({})` `<class 'dict'>`
10. `d = {1:'a', 2:'b', 3:'c'}`
`[d[n] for n in d]` `['a', 'b', 'c']`